Federated Learning on Non-IID Data: Empirical Analysis of Communication Efficiency and Heterogeneity Mitigation

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Abstract

We explore various methods aimed at enhancing efficiency and performance in federated learning systems. Our study encompasses a comparison of FedSGD, FedAvg, and recent extensions to the FedAvg algorithm such as FedProx, FedScaffold, FedGH and FedSAM, aimed at mitigating issues such as client drift. We evaluate these methods on CIFAR-10, analyzing their convergence behavior, communication efficiency, and robustness to non-IID data distributions. Our findings provide agree with existing literature and provide analytical understanding of the trade-offs involved in selecting appropriate federated learning strategies for diverse application scenarios. All our code can be found **here.**

1. Introduction

Federated Learning (FL) enables collaborative model training across decentralized data sources without centralizing sensitive information (McMahan et al., 2017). However, statistical heterogeneity—where clients possess non-identically distributed (non-IID) data—poses fundamental challenges to convergence and model performance (Zhao et al., 2018; Kairouz et al., 2021). Standard Federated Averaging (FedAvg) (McMahan et al., 2017) becomes unstable as local client objectives diverge from the global optimum, particularly when local training epochs increase.

This study addresses two critical aspects of federated optimization: (1) communication efficiency through local computation and client sampling, and (2) heterogeneity mitigation via multiple algorithmic strategies. We systematically investigate four distinct approaches to stabilizing training under non-IID conditions: FedProx (Li et al., 2020) employs proximal regularization to constrain local updates, SCAFFOLD (Karimireddy et al., 2020) uses control variates for variance reduction, gradient harmonization (Zhang et al., 2023) resolves conflicting client updates at the server, and FedSAM (Qu et al., 2022) seeks flatter minima through sharpness-aware optimization. Our contributions include empirical analysis of communication-performance trade-offs and comparative assessment of heterogeneity mitigation techniques across different data distribution scenarios and hyperparameter configurations.

2. Methodology

We conduct experiments on CIFAR-10 with $M = 5$ clients using a convolutional neural network comprising two convolutional layers (32 and 64 filters, 3×3 kernels), max pooling, and two fully connected layers with 128 and 10 units respectively. Training uses SGD with learning rate $\eta = 0.01$, batch size 32. Data is partitioned using IID splits for communication efficiency experiments, ensuring each client receives approximately equal samples across all classes.

2.1. FedSGD vs. Centralized Training

We begin by establishing the theoretical equivalence between Federated Stochastic Gradient Descent (FedSGD) and centralized Stochastic Gradient Descent (SGD). In FedSGD, each client computes gradients based on its local data and sends these gradients to a central server, which aggregates them to update the global model. In contrast, centralized SGD computes gradients using the entire dataset at once. This can be mathematically represented as follows:

$$\begin{aligned} \theta_g^{t+1} &= \theta_g^t - \eta \sum_{i=1}^{M} \frac{N_i}{N} \nabla L_i(\theta_g^t) \\ &= \theta_g^t - \eta \nabla L(\theta_g^t) \end{aligned}$$

where θ_g^t represents the global model parameters at iteration t, η is the learning rate, M is the number of clients, N_i is the number of data points on client i, N is the total number of data points across all clients, and $\nabla L_i(\theta_g^t)$ is the gradient of the loss function computed on client i. The second line follows from the definition of the overall loss function $L(\theta)$ as a weighted sum of local loss functions.

2.1.1. Experimental Setup

We conducted a small experiment using the CIFAR-10 dataset, with five clients, each holding an equal partition of the data. Each client trained a simple toy convolutional neural network (CNN) for 10 SGD steps using both FedSGD and centralized SGD. We ensured that the learning rates, batch sizes, and other hyperparameters were identical across both setups to maintain consistency.

For each iteration t, we recorded the global model parameters θ_g^t from both FedSGD and centralized SGD. In the case of FedSGD, after each client computed its local gradient, we aggregated these gradients on the server to update the global model. In centralized SGD, we computed the gradient using the entire dataset and updated the model accordingly.

As for our evaluation metrics, we calculated the L2 norm difference between the global model parameters obtained via both methods and computed the iteration wise training loss on a held-out validation set after each update.

2.2. Federated Averaging

FedAvg proceeds iteratively: the server broadcasts global model θ_g^t to selected clients, each client performs K local SGD epochs on dataset D_i, yielding updated model $\theta_i^{(K)}$, and the server aggregates via weighted averaging:

$$\theta_g^{t+1} = \sum_{i=1}^{M} \frac{N_i}{N} \theta_i^{(K)} \tag{1}$$

where $N_i = |D_i|$ and $N = \sum_i N_i$.

2.2.1. Varying Local Epochs

We evaluate $K \in \{1, 5, 10, 25\}$ with full client participation to assess communication-computation trade-offs. Weight divergence is measured as:

$$d_\theta^t = \frac{1}{M} \sum_{i=1}^{M} \|\theta_i^{(K)}(t) - \theta_g^t\|_2 \tag{2}$$

2.2.2. Client Sampling

With fixed $K = 1$, we vary client participation fractions $f \in \{1.0, 0.6, 0.4\}$, sampling $\lfloor fM \rfloor$ clients uniformly without replacement per round. This reduces communication overhead proportionally to f.

2.3. Data Heterogeneity Analysis

To study the effects of data heterogenity on FedAvg, we create Dirichlet partitions of CIFAR-10 with concentration parameters $\alpha \in \{100, 1.0, 0.5, 0.1\}$. Lower α induces greater label skew, simulating non-IID conditions. We measure FedAvg performance degradation and divergence escalation as heterogeneity intensifies.

- $\alpha = 100$: This approximates an IID setting where each client has a nearly uniform distribution of all 10 classes.
- $\alpha = 1.0$ and $\alpha = 0.5$: These values represent moderate levels of heterogeneity.
- $\alpha = 0.1$: This creates a highly skewed, non-IID environment where each client's dataset is dominated by a small subset of the classes.

2.3.1. Model and Training

We utilize the same CNN architecture and training hyperparameters as in previous experiments to ensure consistency. Each client trains locally for $K = 5$ epochs per communication round, with full client participation to isolate the effects of data heterogeneity.

2.3.2. Evaluation Metrics

We track test accuracy, training loss, and weight divergence d_θ^t over 50 communication rounds for each α setting. This allows us to quantify how increasing heterogeneity impacts convergence speed and final model quality.

2.4. Mitigating Heterogeneity

2.4.1. FedProx

FedProx (Li et al., 2020) augments the local objective with a proximal term:

$$\min_\theta \mathcal{L}_i(\theta) + \frac{\mu}{2}\|\theta - \theta_g^t\|^2 \tag{3}$$

where $\mu \geq 0$ controls regularization strength. We test $K \in \{1, 5, 10, 25\}$ and $\mu \in \{0.001, 0.01, 0.1\}$ under IID partitioning. The proximal gradient is computed by summing $\mu(\theta - \theta_g^t)$ to standard loss gradients during backpropagation.

2.4.2. SCAFFOLD

SCAFFOLD (Karimireddy et al., 2020) tackles client drift in non-IID federated learning by maintaining *control variates* that explicitly correct for the difference between local and global gradients. The server holds a global control vector c, and each client i maintains its own c_i. When training locally, client i updates its model parameters θ using a corrected gradient of the form

$$g_i^{\text{corr}}(\theta) = \nabla\mathcal{L}_i(\theta) - c_i + c,$$

so that the stochastic update is biased back toward the global descent direction.

At the start of each round, the server broadcasts (θ_g^t, c) to all clients. Each selected client initializes its local copy (θ_i^t, c_i^t) and runs K local epochs of SGD using the corrected gradient $g_i^{\text{corr}}(\theta)$ with learning rate η_{local}. After local training, the client returns both the model difference $\Delta\theta_i$ and the control variate increment Δc_i to the server. The server then performs a weighted average of the model updates and updates c by aggregating the Δc_i as in (Karimireddy et al., 2020).

We implement SCAFFOLD Dirichlet label partitioning with concentration parameter $\alpha = 0.1$ (non-IID setting), and $K = 5$ local epochs for 20 communication rounds, using $mu = 0.001$ for FedProx for comparison.

2.4.3. Gradient Harmonization

Gradient Harmonization (FedGH) addresses client drift in federated learning by detecting and resolving gradient conflicts at the server level. Unlike client-side interventions. FedGH operates entirely at the server aggregation step, making it transparent to clients and compatible with existing federated learning frameworks.

In our implementation of FedGH, the client updates are checked for conflicts during aggregation via negative cosine similarity. For any pair of client updates exhibiting conflict, we project one update onto the normal plane of the other, effectively removing the conflicting component. This harmonized update is then used to update the global model. The process can be summarized as follows:

1. Collect client updates $\Delta\theta_i$ from all clients.
2. For each pair of updates $(\Delta\theta_i, \Delta\theta_j)$, compute the cosine similarity.
3. If the cosine similarity is negative (indicating conflict), project $\Delta\theta_i$ onto the normal plane of $\Delta\theta_j$:

$$\Delta\theta_i' = \Delta\theta_i - \frac{\Delta\theta_i \cdot \Delta\theta_j}{\|\Delta\theta_j\|^2}\Delta\theta_j$$

4. Aggregate the harmonized updates to form the new global model update:

$$\Delta\theta_g = \frac{1}{M}\sum_{i=1}^{M}\Delta\theta_i'$$

We implemented FedGH over 5 clients using the CIFAR-10 dataset, comparing its performance against standard FedAvg. This was done for 50 communication rounds, with each client performing a single local epoch per round.

2.4.4. FedSAM

FedSAM is a federated learning adaptation of Sharpness-Aware Minimization (SAM) (Foret et al., 2021), which aims to utilize the idea that flat minima induce better generalizability to improve model performance across heterogenous clients. By integrating SAM into the local training phase of FL, we expect the models to not only learn from their local data, but also to reside in a uniformly flat minima of the loss landscape, thereby promoting better generalization and alignment with the global aggregated model.

FedSAM modifies the local training process of FedAvg. Instead of using a standard optimizer, clients use SAM. For each batch update, SAM performs a two-step process:

1. **Ascent Step (Perturbation)**: SAM seeks to find the "worst-case" loss in the immediate neighborhood of the current weights w. It does this by computing a perturbation δ that maximizes the loss within a norm-ball of radius ρ.

$$w_{adv} = w + \rho\underbrace{\frac{\nabla L(w)}{\|\nabla L(w)\|}}_{\delta}$$

2. **Descent Step (Update)**: The model weights are then updated by taking a gradient step with respect to the loss computed at the perturbed weights w_{adv}.

$$w \leftarrow w - \eta\nabla L(w_{adv})$$

The rest of the experiment configuration remains consistent with previous setups: 5 clients, CIFAR-10 dataset, 50 communication rounds, and a single local epoch per round. Server-side aggregation is also simply a weighted average of the client models after local training with SAM.

Data heterogeneity is simulated using a Dirichlet distribution with consistent concentration parameter $\alpha = 0.1$ across clients, ensuring a challenging non-IID scenario for evaluating FedSAM's effectiveness.

We study the effect of varying the SAM radius parameter $\rho \in \{0.01, 0.05, 0.1\}$ on the convergence and final performance of the global model.

Test accuracy remains the primary evaluation metric.

3. Results

3.1. FedSGD Equivalence

Table 1 summarizes the overall performance comparison between FedSGD and centralized SGD over 10 iterations. Figures 1 and 2 illustrate the accuracy progression and L2 norm differences between the two methods, respectively.

After 10 iterations, the *average norm difference* between the global model parameters of FedSGD and centralized SGD was found to be **0.004186**, indicating a close alignment between the two methods.

The *average loss difference* on the held-out validation set between FedSGD and centralized SGD across the 10 iterations

Metric	FedSGD	Centralized SGD
Initial Accuracy (iter 1)	8.67%	9.04%
Final Accuracy (iter 10)	9.80%	11.81%
Convergence Pattern	Monotonic	Variable with large gains

Table 1. Comparison of FedSGD and Centralized SGD on CIFAR-10 over 10 iterations

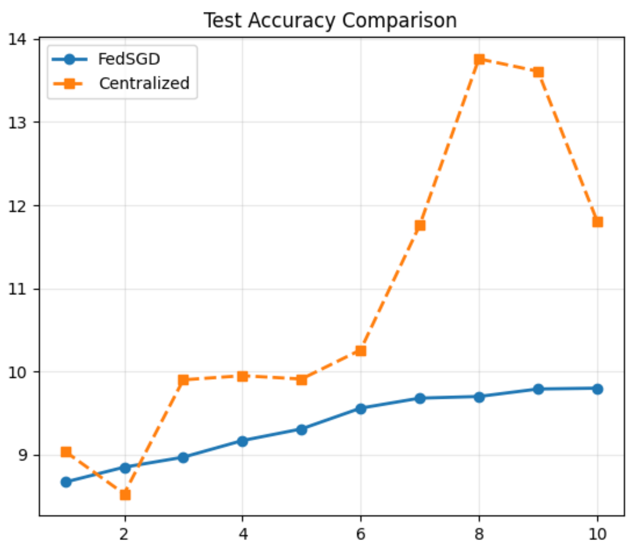


Figure 1. Accuracy Comparison between FedSGD and Centralized SGD over 10 iterations. The y-axis represents accuracy (%) and the x-axis represents the iteration number.

was **0.004057**, further supporting the theoretical equivalence of the two methods.

However, we observed that while FedSGD exhibited a more stable and monotonic convergence pattern, centralized SGD showed variability with occasional large gains in accuracy. This suggests that while both methods are theoretically equivalent, their practical implementations may lead to different convergence behaviors due to factors such as data distribution and gradient aggregation.

3.2. Communication Efficiency Analysis

3.2.1. Impact of Local Epochs

Table 2 summarizes FedAvg performance across local epoch configurations. Test accuracy demonstrates monotonic degradation with increasing K, while final divergence escalates proportionally—exhibiting over threefold amplification between minimal and maximal local epoch settings. Training time exhibits marginal reduction across configurations, indicating communication overhead dominance in the experimental setup.

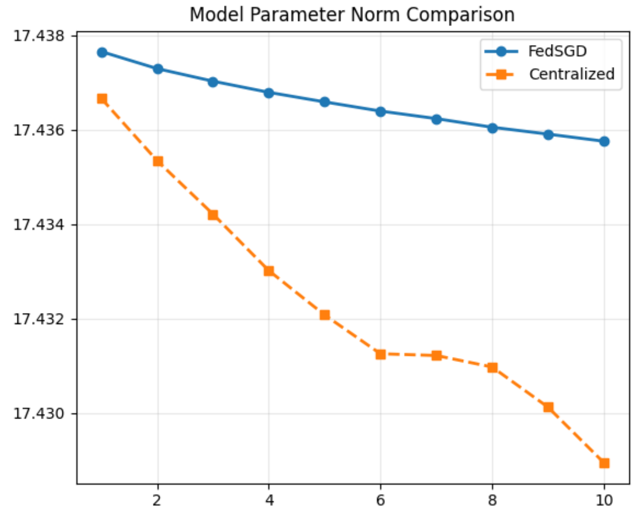


Figure 2. L2 Norm Difference between FedSGD and Centralized SGD model parameters over 10 iterations. The y-axis represents the L2 norm difference and the x-axis represents the iteration number.

K	**Test Acc. (%)**	**Test Loss**	**Divergence**	**Time (min)**
1	78.56	0.633	4.78	13.10
5	78.00	0.651	9.69	12.11
10	76.95	0.680	13.31	11.70
25	75.38	0.741	19.98	11.55

Table 2. FedAvg performance on CIFAR-10 with varying local epochs K.

3.2.2. Client Sampling Strategies

Table 3 presents results under partial client participation. Reducing sampling fraction yields progressive accuracy degradation while substantially decreasing computational cost. Divergence exhibits counterintuitive inverse correlation with participation rate: lower sampling produces reduced aggregate drift metrics.

3.3. Heterogeneity Impact

Table 4 summarizes FedAvg performance across data heterogeneity levels. As α decreases, simulating increased non-IID conditions, final test accuracy degrades significantly from 81.10% at $\alpha = 100$ to 71.89% at $\alpha = 0.1$.

Figure 5a and 5b show how, with higher heterogeneity (lower α), the models converge to worse performing plateaus. Across 50 communication rounds, test accuracy consistently remains lower and test loss higher as α decreases.

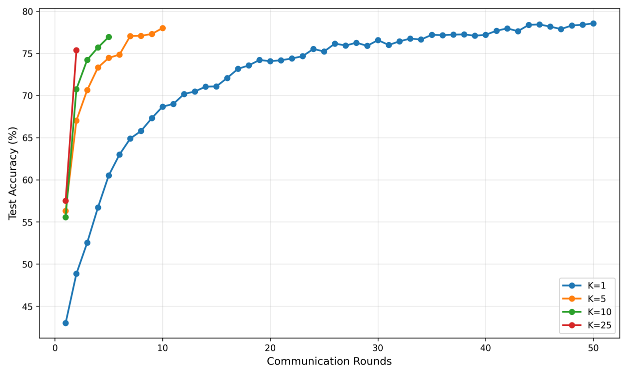


Figure 3. Test accuracy progression for FedAvg with varying local epochs.

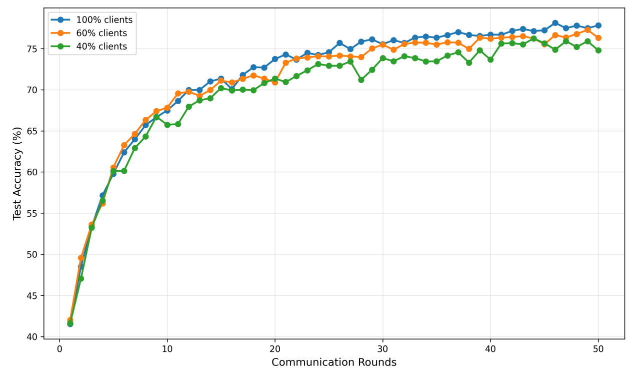


Figure 4. FedAvg accuracy under varying client participation.

Fraction	Test Acc. (%)	Divergence	Time (min)
100%	77.81	4.76	13.00
60%	76.31	4.18	8.36
40%	74.79	3.47	6.01

Table 3. FedAvg under varying client participation fractions.

Dirichlet Alpha (α)	Final Test Acc. (%)
100	81.10
1.0	79.17
0.5	78.71
0.1	71.89

Table 4. FedAvg performance under varying data heterogeneity levels.

Interestingly, the convergence pattern remains qualitatively similar across heterogeneity levels, suggesting that while non-IID data impairs final performance, it does not fundamentally alter the optimization trajectory under FedAvg.

Figure 6 clearly shows that clients drift further apart after 50 communication rounds as data heterogeneity increases. There is almost a 30% increase in final divergence between the most homogeneous ($\alpha = 100$) and most heterogeneous ($\alpha = 0.1$) settings.

3.4. Heterogeneity Mitigation Techniques

3.4.1. FedProx: Proximal Regularization

Table 5 presents FedProx performance across hyperparameter configurations. Optimal performance emerges at maximal local epochs with minimal regularization strength, substantially outperforming baseline FedAvg while achieving dramatic divergence reduction. Excessive regularization demonstrates consistent performance degradation across all local epoch settings, with the highest penalty coefficient yielding results approaching unregularized baseline performance.

The relationship between regularization strength and divergence exhibits strong monotonic correlation: higher μ values progressively constrain local updates. However, this constraint produces diminishing returns beyond the minimal tested value, with intermediate and high regularization configurations showing accuracy deterioration. Test and training losses follow congruent patterns, with optimal hyperparameter choices achieving superior performance on both metrics. The most aggressive local training regime benefits maximally from proximal regularization, validating the technique's utility precisely in high-drift scenarios.

3.4.2. SCAFFOLD: Control Variates

We compare SCAFFOLD (Karimireddy et al., 2020) against FedAvg (McMahan et al., 2017) and FedProx (Li et al., 2020).

Table 6 reports the final test accuracy for the three methods. SCAFFOLD achieves the highest accuracy among the methods, while FedAvg and FedProx obtain similar performance. This pattern is consistent with the control-variate interpretation of SCAFFOLD: by correcting client updates using c and c_i, the method reduces variance and client drift, which translates into better generalization under label skew. At the same time, SCAFFOLD maintains competitive training loss, indicating that its variance reduction does not substantially slow local convergence in this setting.

3.4.3. Gradient Harmonization

Figures 8 and 9 present the test accuracy and test loss comparisons between FedAvg and FedGH over 50 communication rounds. The final test accuracy achieved by FedAvg was 71.27%, while FedGH attained a slightly higher final test accuracy of 71.72%. Though the difference is marginal, it indicates that FedGH's approach to mitigating client drift through gradient harmonization can lead to improved model

Table 5. FedProx results on CIFAR-10 with varying local epochs K and proximal parameter μ under non-IID partitions ($\alpha = 0.1$, 5 clients). Each entry reports the final-round metrics.

K	μ	**Test Acc. (%)**	**Test Loss**	**Train Loss**	**Divergence**
1	0.001	70.59	0.850	0.332	3.59
1	0.01	72.12	0.809	0.361	3.10
1	0.1	65.57	0.984	0.526	1.43
5	0.001	69.18	0.883	0.298	6.72
5	0.01	67.80	0.924	0.412	4.04
5	0.1	52.60	1.327	0.645	1.22
10	0.001	64.76	1.034	0.311	8.75
10	0.01	61.82	1.093	0.463	4.19
10	0.1	41.59	1.593	0.712	1.18
25	0.001	53.10	1.472	0.344	11.62
25	0.01	41.81	1.662	0.546	4.40
25	0.1	32.49	1.861	0.816	1.30

Method	**Final Test Acc. (%)**
FedAvg	65.72
FedProx	65.48
SCAFFOLD	**70.01**

Table 6. FedScaffold, FedProx and FedAvg test accuracy on CIFAR-10 with non-IID partitions ($\alpha = 0.1$), $M = 5$ clients, $K = 5$ local epochs, and 20 rounds. FedProx uses $\mu = 0.001$.

Method	ρ	**Test Acc. (%)**
FedAvg	0.0	72.30
FedSAM	0.01	**72.81**
FedSAM	0.05	72.29
FedSAM	0.1	70.88

Table 7. FedSAM performance on CIFAR-10 with varying SAM radius ρ (non-IID data, $\alpha = 0.1$, 50 rounds).

performance in federated learning settings.

The best accuracy observed for FedAvg was 72.15%, while FedGH achieved a best accuracy of 72.33%. The figures illustrate that both methods achieved similar convergence speeds to a stable loss and accuracy, with FedGH demonstrating a slight advantage in final performance metrics.

Throughout training, FedGH consistently detected and resolved between 2-8 gradient conflicts per communication round, with the frequency generally increasing; suggesting the presence of data heterogeneity among clients.

3.4.4. FedSAM: Sharpness-Aware Minimization

Table 7 summarizes FedSAM performance across varying SAM radius ρ values. Optimal performance occurs at $\rho = 0.01$, surpassing baseline FedAvg accuracy by 0.51%. Both smaller and larger radius settings yield diminished accuracy, indicating sensitivity to this hyperparameter.

Figure 10a shows test accuracy progression for different ρ values over 50 communication rounds. The $\rho = 0.01$ configuration consistently outperforms other chosen values, and overtakes the performance of FedAvg after approximately 30 rounds.

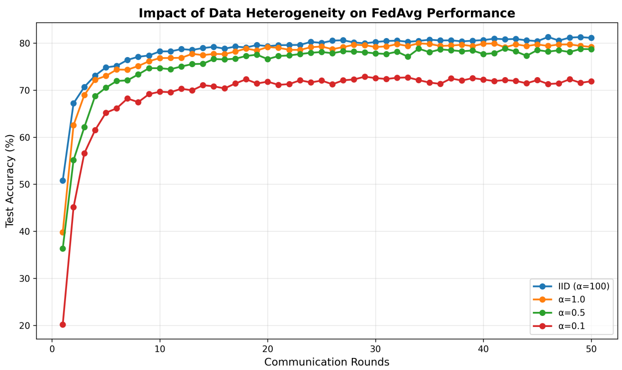
We see an interesting observation in Figure 10b, where the flatter minima induced by SAM actually also result in an overall lower loss that the global model converges to. This suggests that the sharpness-aware updates not only help with generalization but also guide the optimization towards more optimal regions of the loss landscape.

Client weight divergence trends in Figure 11 reveal that larger ρ values induce greater divergence from the global model. This aligns with the intuition that more aggressive sharpness-aware updates lead to more pronounced local model deviations, which can be detrimental in highly heterogeneous settings. It is imperative to note, however, that higher divergence does not necessarily correlate with worse performance, as evidenced by the optimal accuracy achieved at $\rho = 0.01$ despite moderate divergence levels.

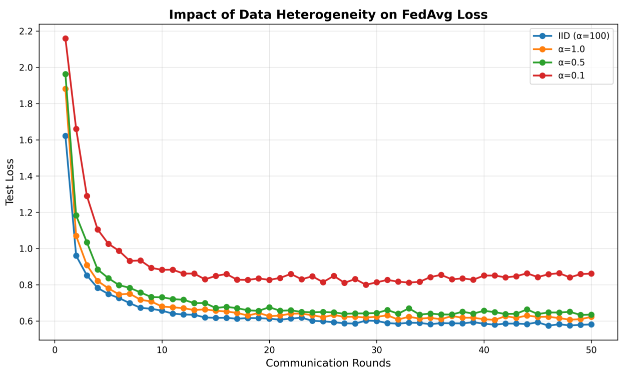
4. Discussion

4.1. Theoretical Equivalence of FedSGD and Centralized SGD

We observed non-trivial differences in convergence behaviour between FedSGD and centralized SGD, despite their theoretical equivalence. Figures 1 and 2 illustrate that



(a) Test Accuracy under varying data heterogeneity levels.



(b) Test Loss under varying data heterogeneity levels.

Figure 5. FedAvg performance metrics across different Dirichlet alpha values.

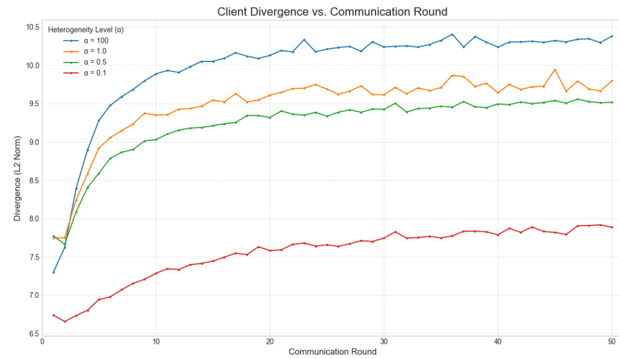


Figure 6. Weight Divergence under varying data heterogeneity levels.

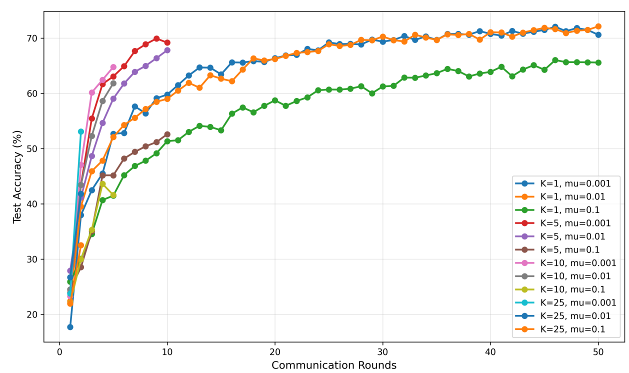


Figure 7. FedProx accuracy convergence comparison for varying values of K and mu.

while both methods converge to similar accuracy levels and model parameters, the paths they take differ significantly, with an increasing divergence trend in later iterations.

The fundamental issue is that FedSGD and centralized SGD process different data in each iteration. In FedSGD, each client computes gradients based on its local data, which may not represent the overall data distribution, leading to biased gradient estimates. Even with IID data, the batches processed by each client may differ from those in centralized SGD. Moreover, momentum can exacerbate these differences by accumulating past gradients, amplifying discrepancies over time (figure 2).

Our results highlight the importance of considering practical implementation details when deploying federated learning algorithms. While theoretical equivalence provides a useful framework, real-world factors such as data distribution and optimization techniques can lead to significant differences in performance and convergence behaviour.

4.2. FedAvg Communication–Computation trade-offs

The experiments probe the classic trade-off in federated optimization: increasing local computation per round reduces synchronization frequency but risks misalignment between local and global objectives (Table 2). In our FedAvg runs this appears as degraded global progress (Figure 3) and higher measured divergence when K is large.

Client sampling interacts with this effect: selecting fewer clients injects stochasticity that can blunt persistent directional drift but reduces information available to the server (Figure 4). The practical result is a cost–benefit tension—sampling lowers per-round communication yet often requires more rounds to reach the same global performance (Table 3). The divergence metric succinctly captures these dynamics: higher divergence signals greater local desynchronization and helps explain why aggressive local work can hurt global accuracy even when local losses improve (McMahan et al., 2017; Zhao et al., 2018).

4.3. Data Heterogeneity Effects

The observed performance degradation with increasing data heterogeneity aligns with established literature (Zhao et al., 2018; Kairouz et al., 2021). As the Dirichlet concentration parameter α decreases, clients' data distributions become more skewed, leading to significant "client drift" during lo-

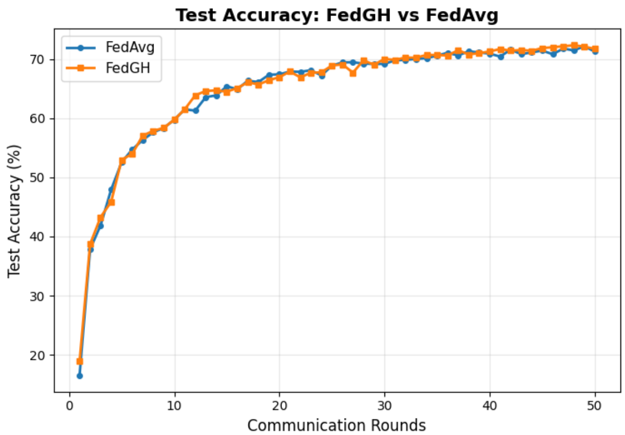


Figure 8. Test Accuracy Comparison between FedAvg and FedGH over 50 communication rounds. The accuracy for both methods improves consistently over rounds, with FedGH showing a slight edge in middle and later rounds.

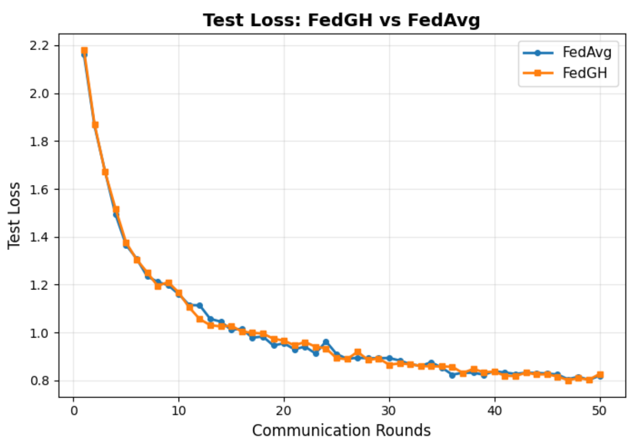
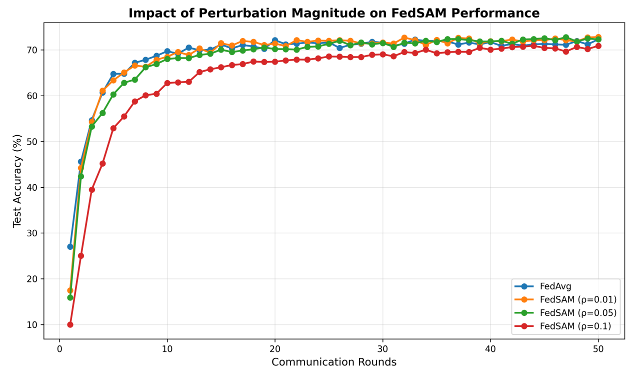
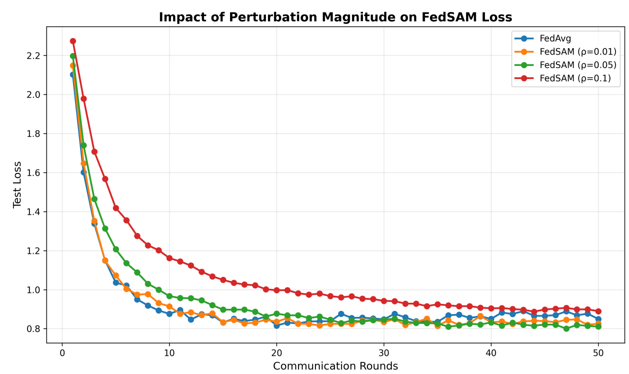


Figure 9. Test Loss Comparison between FedAvg and FedGH over 50 communication rounds. The loss for both methods follow similar trends throughout the training process.



(a) Test Accuracy of FedSAM with varying ρ values over 50 communication rounds.



(b) Test Loss of FedSAM with varying ρ values over 50 communication rounds.

Figure 10. FedSAM performance metrics across different SAM radius values.

cal training (Figure 6). The local models move in divergent directions, and when averaged, the resulting global model is suboptimal. In extreme cases, conflicting updates can cancel each other out, leading to stagnation or oscillation. These results suggest that, as data distribution deviates from I.I.D., the effectiveness of FedAvg diminishes.

4.4. Heterogeneity Mitigation Techniques

4.4.1. FedProx: Proximal Regularization

The FedProx results in Table 5 illustrate the classic bias–variance trade-off introduced by proximal regularization (Li et al., 2020). For a fixed K, increasing μ consistently reduces divergence but comes at the cost of higher losses and lower test accuracy, especially for large μ. Across different values of K at small or moderate μ, FedProx still exhibits growing divergence as local training becomes more aggressive, indicating that the proximal term alone does not fully offset misalignment when K is large. Instead, relatively small values of μ with modest K offer the best balance between stability and performance.

These findings are consistent with the original motivation for FedProx (Li et al., 2020; Kairouz et al., 2021): the proximal term is most useful as a gentle corrective mechanism rather than a cure-all for extreme client drift. In practice, monitoring divergence and adjusting μ only when drift becomes problematic may provide a more robust deployment strategy than fixing a large proximal weight a priori.

4.4.2. SCAFFOLD under non-IID data

The comparison in Table 6 shows that SCAFFOLD consistently outperforms both FedAvg and FedProx in the non-IID scenario with $K = 5$ local epochs. While FedAvg and FedProx achieve similar final accuracies, SCAFFOLD attains a noticeably higher value with lower test loss. This behaviour aligns with the control-variate perspective in (Karimireddy

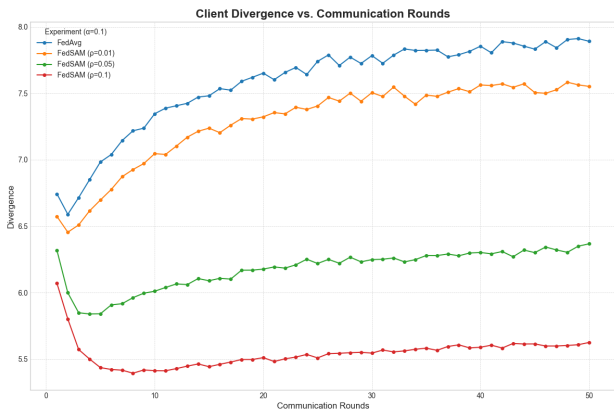


Figure 11. Weight Divergence of FedSAM with varying ρ values over 50 communication rounds.

Method	**Final Test Acc. (%)**
FedAvg (baseline)	71.27
FedProx ($\mu = 0.001$)	70.59
FedGH	71.72
FedSAM ($\rho = 0.01$)	72.81
SCAFFOLD*	70.01
*20 rounds; others 50 rounds	

Table 8. Performance comparison under severe non-IID conditions ($\alpha = 0.1$, $K = 5$ local epochs, CIFAR-10).

et al., 2020). By explicitly correcting local gradients toward a global direction through $g_i^{\text{corr}}(\theta) = \nabla\mathcal{L}_i(\theta) - c_i + c$, SCAFFOLD mitigates the effect of heterogeneous client objectives without relying solely on shrinking local steps or adding proximal penalties. Compared to FedProx (Li et al., 2020), SCAFFOLD thus trades a modest increase in algorithmic complexity and communication for more reliable performance under label skew (Kairouz et al., 2021).

4.4.3. Gradient Harmonization - FedGH

Our experimental comparison between FedAvg and FedGH highlighted the potential of gradient harmonization to mitigate client drift. Though modest, the improved performance suggests that addressing gradient conflicts during aggregation can enhance model convergence, particularly in non-IID settings. FedGH maintained stable training dynamics similar to FedAvg, indicating that the harmonization process did not introduce significant instability.

The implementation of FedGH is relatively simple, requiring only modifications at the server aggregation step, making it attractive for practitioners without overhauling client-side protocols. However, the $\mathcal{O}(M^2)$ complexity becomes a bottleneck when scaling to larger numbers of clients. While easy to implement, FedGH's reliance on cosine similarity for conflict detection may not capture all forms of gradient divergence, especially in high-dimensional parameter spaces. This method is a practical, lightweight approach to addressing client drift, particularly if combined with other techniques such as FedProx or SCAFFOLD.

4.4.4. FedSAM - Sharpness-Aware Minimization

Our results show that applying domain generalization techniques, particularly Sharpness-Aware Minimization (SAM) in pursuit of flatter minima, can yield better performance in FL contexts under careful tuning of ρ. This suggests that data heterogeneity amongst clients is a problem similar to domain shift, and techniques that improve generalization across domains can help mitigate the negative effects of non-IID data.

However, the sensitivity of FedSAM to ρ indicates a delicate balance between promoting flatness and maintaining effective local learning. A small radius ($\rho = 0.01$) provided enough perturbation to guide models towards flatter regions without excessively disrupting local optimization. In contrast, larger radii ($\rho = 0.05$ and $\rho = 0.1$) likely introduced too much perturbation, causing local models to diverge significantly from their optimal points, which hurt overall convergence and accuracy.

4.5. Unified Comparison of Heterogeneity Mitigation Techniques

Table 8 consolidates performance metrics across all evaluated methods under severe non-IID conditions ($\alpha = 0.1$, $K = 5$ local epochs).

4.5.1. Comparative Analysis and Practical Recommendations

FedSAM achieved the highest accuracy through sharpness-aware optimization that seeks flatter minima, effectively treating heterogeneity as a domain shift problem (Foret et al., 2021; Qu et al., 2022), but required doubled local computation. SCAFFOLD demonstrated fastest early convergence through variance reduction (Karimireddy et al., 2020), reaching target accuracies in roughly half the rounds despite doubled communication overhead. FedGH provided marginal improvement with zero client overhead but faces $O(M^2)$ scaling limitations beyond small federations. FedProx unexpectedly underperformed despite its theoretical motivation, with optimal μ depending critically on the specific combination of K, α, and client characteristics (Li et al., 2020).

Method selection should align with deployment constraints. For accuracy-critical applications with adequate compute, FedSAM offers the best final performance. In communication-limited settings, SCAFFOLD's rapid convergence may justify its doubled per-round cost—reaching targets in 10 rounds versus 20+ for FedAvg represents net savings. For large-scale federations ($M > 50$), FedProx remains the only scalable option, as FedGH's quadratic scaling fails and SCAFFOLD's state management complicates client dropout. FedGH suits rapid deployment in small federations through server-only modifications.

Three key insights emerge: variance reduction (SCAFFOLD) proves more effective than regularization (FedProx) under severe heterogeneity by directly correcting gradient bias; methods addressing different aspects are orthogonal and potentially composable; and all methods exhibit hyperparameter sensitivity dependent on heterogeneity characteristics. Limitations include small federation size ($M = 5$) not reflecting real-world partial participation, focus on label skew rather than other heterogeneity types, and limited hyperparameter exploration. Future work should investigate method combinations, larger-scale federations, and adaptive hyperparameter scheduling for improved robustness (Kairouz et al., 2021).

5. Conclusion

This work empirically investigates federated learning under communication constraints and data heterogeneity. We verified the theoretical equivalence between FedSGD and centralized SGD, then demonstrated fundamental trade-offs in FedAvg: increasing local epochs reduces communication but degrades accuracy and amplifies client divergence. Under severe data heterogeneity, vanilla FedAvg performance deteriorates significantly, motivating our evaluation of mitigation strategies.

Among heterogeneity techniques, FedSAM achieved the highest accuracy through sharpness-aware optimization but required doubled local computation. SCAFFOLD converged fastest via variance reduction despite doubled communication overhead. FedGH provided modest improvements with zero client cost but poor scalability, while FedProx surprisingly underperformed due to hyperparameter sensitivity. These findings demonstrate that effective federated optimization requires matching algorithmic choice to deployment constraints: SCAFFOLD for bandwidth-rich scenarios prioritizing speed, FedSAM when computation is tolerable, and FedGH for lightweight improvements in small federations. The observed sensitivity across all methods highlights the need for adaptive mechanisms as federated learning scales to production deployments with diverse client populations.

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Appendix

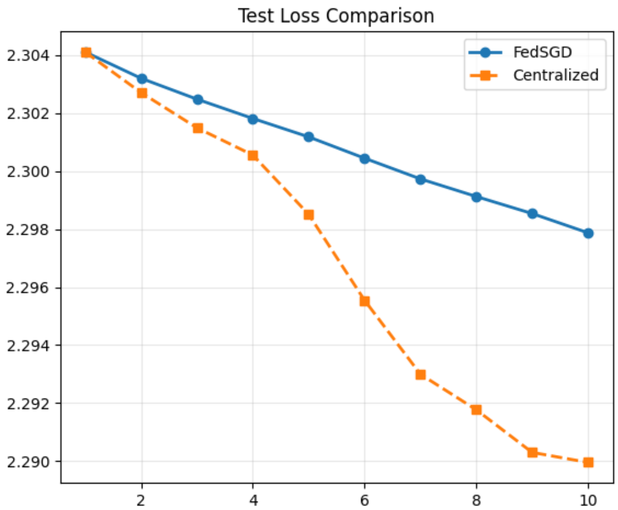

Figure 12. Loss Comparison between FedSGD and Centralized SGD over 10 iterations. The y-axis represents the loss value and the x-axis represents the iteration number. Observe how the loss values begin to diverge in later iterations.

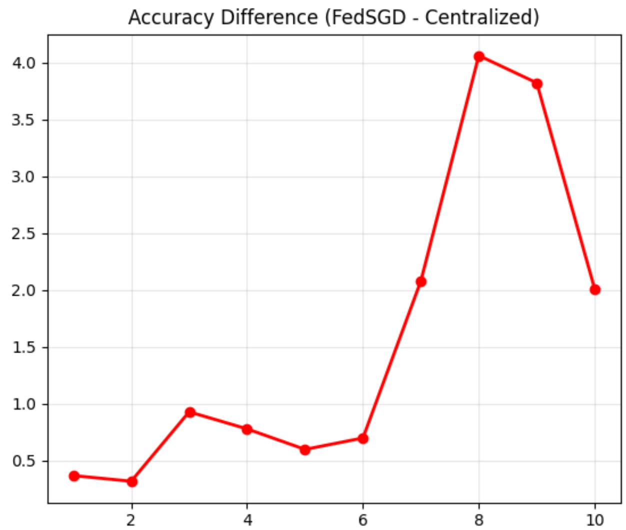

Figure 13. Accuracy Difference between FedSGD and Centralized SGD over 10 iterations. The y-axis represents the accuracy difference (%) and the x-axis represents the iteration number. The accuracy difference tends to increase in later iterations, indicating divergence in convergence behaviour.

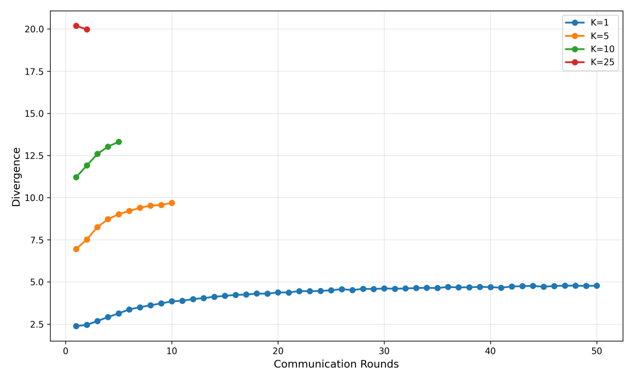

Figure 14. FedAvg divergence progression with varying local epochs K.

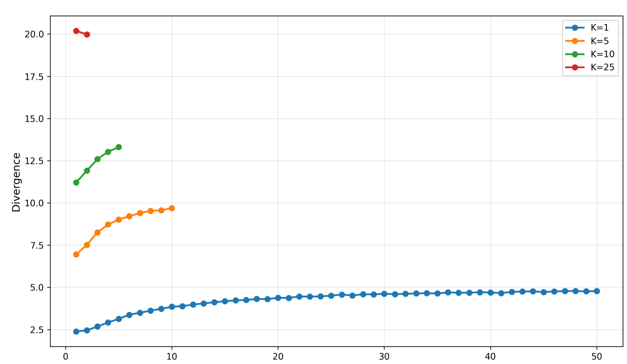

Figure 15. FedAvg loss progression with varying local epochs K.

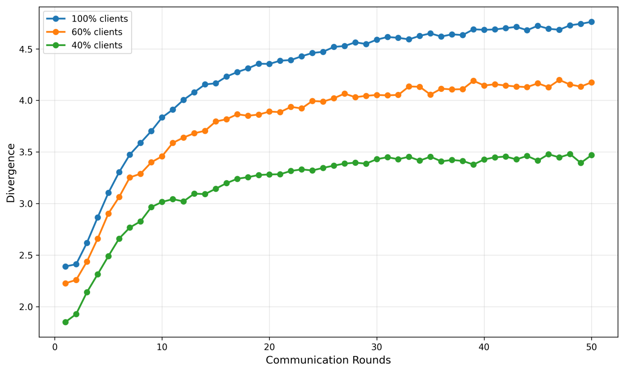

Figure 16. FedAvg divergence under different client participation fractions.

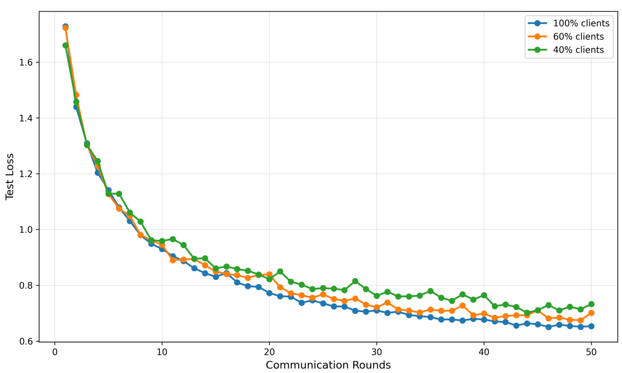

Figure 17. FedAvg loss under different client participation fractions.

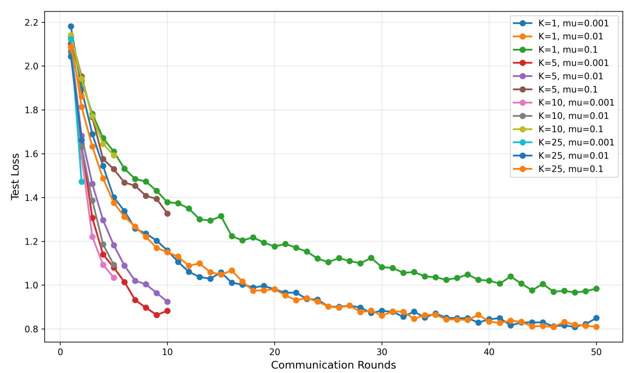

Figure 19. FedProx loss.

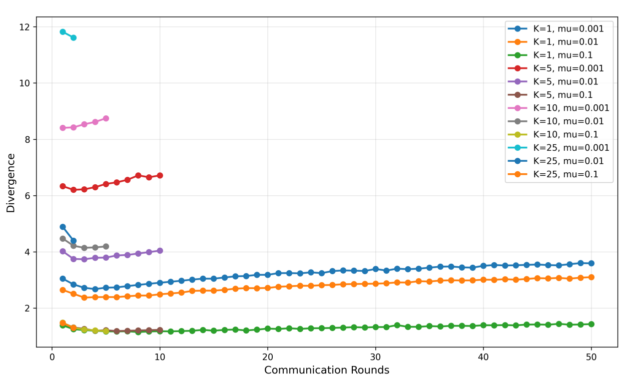

Figure 18. FedProx divergence.